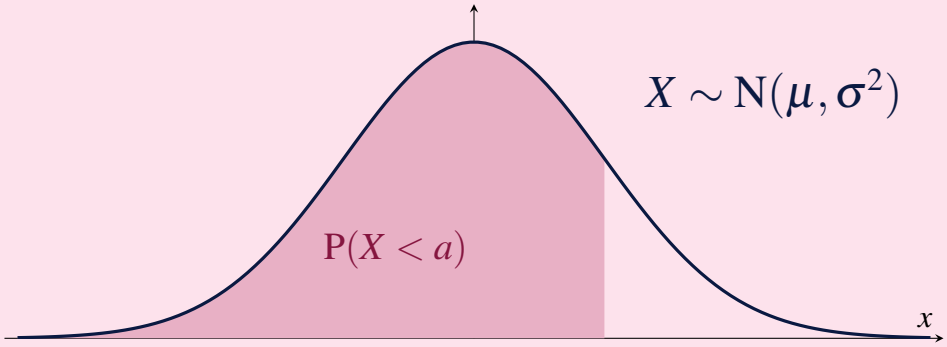
Probability Distributions

IB Mathematics AA SL · Past-Paper Questions

Discrete random variables & expected value · Binomial distribution · Normal distribution



How to use this booklet

- Read the pink **Key Notes** pages first.
- 4 sections: discrete random variables & expected value, binomial distribution, normal distribution, and mixed normal & binomial.
- Inside each section the questions run from the **most recent exam to the oldest**; the exam session, paper and question number are shown above every question.
- Each white **Question** page is followed by a yellow **Answer** page with full working.
- Paper 1 = no calculator; Paper 2 = calculator (GDC) allowed.

Key Notes

1. Discrete random variables

- A probability distribution lists every value x and its probability $\mathrm{P}(X = x)$.
- All probabilities lie between 0 and 1 and **add up to 1**: $\sum \mathrm{P}(X = x) = 1$. Use this to find an unknown such as k or p.
- Probabilities may be written as expressions (e.g. $2k^2$, $\cos\theta$): solve the equation, then **reject** answers that give a probability outside $[0, 1]$.
- "At least" and "at most" questions: add the relevant probabilities, or use $1 -$ P(complement).

2. Expected value $\mathrm{E}(X)$

$$\mathrm{E}(X) = \sum x\mathrm{P}(X = x) \qquad \text{(multiply each value by its probability and add)}$$

- $\mathrm{E}(X)$ is the long-run mean value. Expected number of times $= n \times p$.
- **Fair game**: expected gain $= 0$, i.e. expected winnings $=$ cost to play.
- If $\mathrm{E}(X)$ is a quadratic in p, its maximum is at the vertex $p = -\frac{b}{2a}$ (check all probabilities stay in $[0, 1]$).
- Conditional: $\mathrm{P}(A \mid B) = \dfrac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$, e.g. $\mathrm{P}(X = 2 \mid X > 0) = \dfrac{\mathrm{P}(X = 2)}{\mathrm{P}(X > 0)}$.

3. Binomial distribution $X \sim \mathrm{B}(n, p)$

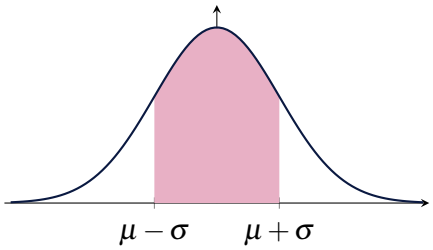
Use when there are n **independent** trials, each with only two outcomes and the **same** probability p of success.

$$\mathrm{P}(X = r) = \binom{n}{r} p^r (1-p)^{n-r} \qquad \mathrm{E}(X) = np \qquad \mathrm{Var}(X) = np(1-p)$$

- GDC: **binompdf**(n, p, r) for $\mathrm{P}(X = r)$; **binomcdf**(n, p, r) for $\mathrm{P}(X \leq r)$.
- $\mathrm{P}(X < r) = \mathrm{P}(X \leq r-1)$, $\mathrm{P}(X \geq r) = 1 - \mathrm{P}(X \leq r-1)$, $\mathrm{P}(X \geq 1) = 1 - (1-p)^n$.
- "The kth success on the nth trial": $(k-1)$ successes in the first $n-1$ trials, **then** a success.

4. Normal distribution $X \sim \mathrm{N}(\mu, \sigma^2)$

- Bell-shaped and symmetric about the mean μ; total area $= 1$.
- $\mathrm{P}(X < \mu) = \mathrm{P}(X > \mu) = 0.5$.
- About 68% within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, 99.7% within $\mu \pm 3\sigma$.
- For a continuous variable $\mathrm{P}(X < a) = \mathrm{P}(X \leq a)$.



5. Standardising — the z-score

$$z = \frac{x - \mu}{\sigma} \qquad Z \sim \mathrm{N}(0,1) \qquad x = \mu + z\sigma$$

- GDC: **normalcdf**(lower, upper, μ, σ) gives a probability.
- **invNorm**(area to the left, μ, σ) gives the value x (or z if $\mu = 0, \sigma = 1$).
- Paper 1 (no GDC): use symmetry, e.g. $\mathrm{P}(93 < X < 107) = 2\,\mathrm{P}(100 < X < 107)$ when $\mu = 100$.

6. Finding an unknown μ or σ

1. Change the given probability into a z-value with invNorm (area to the **left**).
2. Write $\dfrac{x - \mu}{\sigma} = z$.
3. One unknown: rearrange. Two unknowns: two equations, solve **simultaneously**.

e.g. $\mathrm{P}(X < 5) = 0.04$, $\mathrm{P}(X < 25) = 0.7$: $5 = \mu - 1.7507\sigma$, $25 = \mu + 0.5244\sigma$.

7. Normal & binomial together

- Step 1: use the normal distribution to find $p = \mathrm{P}(\text{one item has the property})$.
- Step 2: for n independent items, $Y \sim \mathrm{B}(n, p)$ — then find $\mathrm{P}(Y = r)$, $\mathrm{P}(Y \geq r)$ or $\mathrm{E}(Y) = np$.
- Conditional: $\mathrm{P}(X < b \mid X > a) = \dfrac{\mathrm{P}(a < X < b)}{\mathrm{P}(X > a)}$.

1

May 2021 · Paper 1 · TZ1 · Q9

[Maximum mark: 16]

A biased four-sided die, A, is rolled. Let X be the score obtained when die A is rolled. The probability distribution for X is given in the following table.

x	1	2	3	4
$\mathrm{P}(X=x)$	p	p	p	$\frac{1}{2}p$

(a) Find the value of p. [2]

(b) Hence, find the value of $\mathrm{E}(X)$. [2]

A second biased four-sided die, B, is rolled. Let Y be the score obtained when die B is rolled. The probability distribution for Y is given in the following table.

y	1	2	3	4
$\mathrm{P}(Y=y)$	q	q	q	r

(c) (i) State the range of possible values of r.

(ii) Hence, find the range of possible values of q. [3]

(d) Hence, find the range of possible values for $\mathrm{E}(Y)$. [3]

Agnes and Barbara play a game using these dice. Agnes rolls die A once and Barbara rolls die B once. The probability that Agnes' score is less than Barbara's score is $\frac{1}{2}$.

(e) Find the value of $\mathrm{E}(Y)$. [6]

1 *May 2021 · Paper 1 · TZ1 · Q9*

(a) $3p+\frac{1}{2}p=1 \Rightarrow \boxed{p=\frac{2}{7}}$ (b) $\mathrm{E}(X)=\frac{2}{7}(1+2+3)+4\cdot\frac{1}{7}=\boxed{\frac{16}{7}}$

(c)(i) $\boxed{0\le r\le 1}$ (ii) $3q+r=1 \Rightarrow q=\frac{1-r}{3}$: $\boxed{0\le q\le\frac{1}{3}}$

(d) $\mathrm{E}(Y)=6q+4r=6q+4(1-3q)=4-6q$, so $\boxed{2\le \mathrm{E}(Y)\le 4}$

(e) $\mathrm{P}(X<Y)=\frac{2}{7}(2q+r)+\frac{2}{7}(q+r)+\frac{2}{7}r=\frac{6}{7}(q+r)=\frac{1}{2}\Rightarrow q+r=\frac{7}{12}$.
With $3q+r=1$: $q=\frac{5}{24}$, $r=\frac{3}{8}$. $\mathrm{E}(Y)=4-6\left(\frac{5}{24}\right)=\boxed{\frac{11}{4}}$

2

May 2021 · Paper 2 · TZ2 · Q9

[Maximum mark: 15]

All answers in this question should be given to four significant figures.

In a local weekly lottery, tickets cost $\$2$ each.

In the first week of the lottery, a player will receive $\$D$ for each ticket, with the probability distribution shown in the following table. For example, the probability of a player receiving $\$10$ is 0.03. The grand prize in the first week of the lottery is $\$1000$.

d	0	2	10	50	Grand Prize
$\mathrm{P}(D=d)$	0.85	c	0.03	0.002	0.0001

(a) Find the value of c. [2]

(b) Determine whether this lottery is a fair game in the first week. Justify your answer. [4]

If nobody wins the grand prize in the first week, the probabilities will remain the same, but the value of the grand prize will be $\$2000$ in the second week, and the value of the grand prize will continue to double each week until it is won. All other prize amounts will remain the same.

(c) Given that the grand prize is not won and the grand prize continues to double, write an expression in terms of n for the value of the grand prize in the nth week of the lottery. [2]

The wth week is the first week in which the player is expected to make a profit. Ryan knows that if he buys a lottery ticket in the wth week, his expected profit is $\$p$.

(d) Find the value of p. [7]

2 *May 2021 · Paper 2 · TZ2 · Q9*

(a) $c = 1 - (0.85 + 0.03 + 0.002 + 0.0001) = \boxed{0.1179}$

(b) $\mathrm{E}(D) = 2(0.1179) + 10(0.03) + 50(0.002) + 1000(0.0001) = 0.7358$. Expected gain $= 0.7358 - 2 = -1.264 \neq 0$, so **not fair**.

(c) $\boxed{1000 \times 2^{n-1}}$

(d) Profit if $0.6358 + 0.0001 \times 1000 \times 2^{w-1} > 2 \Rightarrow 2^{w-1} > 13.642 \Rightarrow w = 5$.

$p = 0.6358 + 0.0001(16000) - 2 = \boxed{0.2358}$

3

November 2020 · Paper 2 · Q3

A discrete random variable X has the following probability distribution.

x	0	1	2	3
$\mathrm{P}(X=x)$	q	$4p^2$	p	$0.7-4p^2$

(a) Find an expression for q in terms of p. [2]

(b) (i) Find the value of p which gives the largest value of $\mathrm{E}(X)$.

(ii) Hence, find the largest value of $\mathrm{E}(X)$. [4]

3 *November 2020 · Paper 2 · Q3*

(a) $q+4p^2+p+0.7-4p^2=1 \Rightarrow \boxed{q=0.3-p}$

(b)(i) $\mathrm{E}(X)=4p^2+2p+3(0.7-4p^2)=-8p^2+2p+2.1$; max when $-16p+2=0$: $\boxed{p=\frac{1}{8}}$ (ii) $\boxed{\mathrm{E}(X)=2.225}$

4 **November 2019 · Paper 2 · Q7**

[Maximum mark: 7]

The following table shows the probability distribution of a discrete random variable X, where $a \geq 0$ and $b \geq 0$.

x	1	4	a	$a + b - 0.5$
$P(X = x)$	0.2	0.5	b	a

(a) Show that $b = 0.3 - a$. [1]

(b) Find the difference between the greatest possible expected value and the least possible expected value. [6]

4 *November 2019 · Paper 2 · Q7*

(a) $0.2+0.5+b+a=1 \Rightarrow b=0.3-a$ ✓ (so the last x-value is $a+b-0.5=-0.2$)

(b) $\mathrm{E}(X)=0.2+2+ab-0.2a=2.2+0.1a-a^2,\ 0\le a\le 0.3.$

Max at $a=0.05$: 2.2025; min at $a=0.3$: 2.14. Difference = $\boxed{0.0625}$

5

May 2019 · Paper 2 · TZ1 · Q10

There are three fair six-sided dice. Each die has two green faces, two yellow faces and two red faces.

All three dice are rolled.

(a) (i) Find the probability of rolling exactly one red face.

(ii) Find the probability of rolling two or more red faces. [5]

Ted plays a game using these dice. The rules are:

- Having a turn means to roll all three dice.
- He wins $\$10$ for each green face rolled and adds this to his winnings.
- After a turn Ted can either:
  - end the game (and keep his winnings), or
  - have another turn (and try to increase his winnings).
- If two or more red faces are rolled in a turn, all winnings are lost and the game ends.

(b) Show that, after a turn, the probability that Ted adds exactly $\$10$ to his winnings is $\frac{1}{3}$. [5]

The random variable D ($\$$) represents how much is added to his winnings after a turn.

The following table shows the distribution for D, where $\$w$ represents his winnings in the game so far.

D ($\$$)	$-w$	0	10	20	30
$\mathrm{P}(D=d)$	x	y	$\frac{1}{3}$	$\frac{2}{9}$	$\frac{1}{27}$

(c) (i) Write down the value of x.

(ii) Hence, find the value of y. [3]

Ted will always have another turn if he expects an increase to his winnings.

(d) Find the least value of w for which Ted should end the game instead of having another turn. [3]

5 *May 2019 · Paper 2 · TZ1 · Q10*

(a)(i) $3\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^2 = \boxed{\frac{4}{9}}$ (ii) $1-\left(\frac{2}{3}\right)^3-\frac{4}{9} = \boxed{\frac{7}{27}}$

(b) Exactly one green and at most one red: $3\cdot\frac{1}{3}\cdot\left[\left(\frac{1}{3}\right)^2+2\left(\frac{1}{3}\right)^2\right]=\frac{1}{3}$ ✓

(c)(i) $x = \mathrm{P}(\geq 2 \text{ red}) = \boxed{\frac{7}{27}}$ (ii) $y = 1-\frac{7}{27}-\frac{1}{3}-\frac{2}{9}-\frac{1}{27} = \boxed{\frac{4}{27}}$

(d) Stop when $\mathrm{E}(D)\leq 0$: $-\frac{7w}{27}+\frac{10}{3}+\frac{40}{9}+\frac{30}{27}\leq 0 \Rightarrow w \geq 34.3$; winnings are multiples of 10: $\boxed{w=40}$

6

May 2019 · Paper 1 · TZ2 · Q1

The following table shows the probability distribution of a discrete random variable X.

X	0	1	2	3
$\mathrm{P}(X=x)$	$\frac{3}{13}$	$\frac{1}{13}$	$\frac{4}{13}$	k

(a) Find the value of k. [3]

(b) Find $\mathrm{E}(X)$. [3]

6 *May 2019 · Paper 1 · TZ2 · Q1*

(a) $\frac{3+1+4}{13} + k = 1 \Rightarrow \boxed{k = \frac{5}{13}}$ (b) $\mathrm{E}(X) = \frac{0+1+8+15}{13} = \boxed{\frac{24}{13}}$

7 **November 2018 · Paper 1 · Q9**

A bag contains n marbles, two of which are blue. Hayley plays a game in which she randomly draws marbles out of the bag, one after another, without replacement. The game ends when Hayley draws a blue marble.

(a) Find the probability, in terms of n, that the game will end on her

(i) first draw;

(ii) second draw. [4]

(b) Let $n = 5$. Find the probability that the game will end on her

(i) third draw;

(ii) fourth draw. [4]

Hayley plays the game when $n = 5$. She pays $\$20$ to play and can earn money back depending on the number of draws it takes to obtain a blue marble. She earns no money back if she obtains a blue marble on her first draw. Let M be the amount of money that she earns back playing the game. This information is shown in the following table.

Number of draws	1	2	3	4
Money earned back ($\$M$)	0	20	$8k$	$12k$

(c) Find the value of k so that this is a fair game. [7]

7 *November 2018 · Paper 1 · Q9*

(a)(i) $\boxed{\frac{2}{n}}$ (ii) $\boxed{\frac{n-2}{n} \times \frac{2}{n-1}}$

(b)(i) $\frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} = \boxed{\frac{1}{5}}$ (ii) $\frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} \cdot 1 = \boxed{\frac{1}{10}}$

(c) $\mathrm{E}(M) = 0\left(\frac{2}{5}\right) + 20\left(\frac{3}{10}\right) + 8k\left(\frac{1}{5}\right) + 12k\left(\frac{1}{10}\right) = 6 + 2.8k$. Fair: $6 + 2.8k = 20 \Rightarrow \boxed{k = 5}$

8

May 2018 · Paper 2 · TZ1 · Q2

A biased four-sided die is rolled. The following table gives the probability of each score.

Score	1	2	3	4
Probability	0.28	k	0.15	0.3

(a) Find the value of k. [2]

(b) Calculate the expected value of the score. [2]

(c) The die is rolled 80 times. On how many rolls would you expect to obtain a three? [2]

9

November 2017 · Paper 2 · Q4

[Maximum mark: 8]

A discrete random variable X has the following probability distribution.

X	0	1	2	3
$\mathrm{P}(X=x)$	0.475	$2k^2$	$\frac{k}{10}$	$6k^2$

(a) Find the value of k. [4]

(b) Write down $\mathrm{P}(X=2)$. [1]

(c) Find $\mathrm{P}(X=2 \mid X>0)$. [3]

8 *May 2018 · Paper 2 · TZ1 · Q2*

(a) $k = 1 - 0.28 - 0.15 - 0.3 = \boxed{0.27}$ (b) $0.28 + 0.54 + 0.45 + 1.2 = \boxed{2.47}$ (c) $80 \times 0.15 = \boxed{12}$

9 *November 2017 · Paper 2 · Q4*

(a) $0.475 + 8k^2 + 0.1k = 1 \Rightarrow 8k^2 + 0.1k - 0.525 = 0 \Rightarrow \boxed{k = 0.25}$ (b) $\frac{0.25}{10} = \boxed{0.025}$

(c) $\dfrac{0.025}{1 - 0.475} = \boxed{\frac{1}{21}\ (0.0476)}$

10

May 2017 · Paper 1 · TZ1 · Q10

The following table shows the probability distribution of a discrete random variable A, in terms of an angle θ.

a	1	2
$\mathrm{P}(A = a)$	$\cos\theta$	$2\cos 2\theta$

(a) Show that $\cos\theta = \frac{3}{4}$. [6]

(b) Given that $\tan\theta > 0$, find $\tan\theta$. [3]

(c) Let $y = \frac{1}{\cos x}$, for $0 < x < \frac{\pi}{2}$. The graph of y between $x = \theta$ and $x = \frac{\pi}{4}$ is rotated 360° about the x-axis. Find the volume of the solid formed. [6]

10 *May 2017 · Paper 1 · TZ1 · Q10*

(a) $\cos\theta + 2(2\cos^2\theta - 1) = 1 \Rightarrow 4\cos^2\theta + \cos\theta - 3 = 0 \Rightarrow (4\cos\theta - 3)(\cos\theta + 1) = 0$; probability ≥ 0, so $\cos\theta = \frac{3}{4}$ ✓

(b) $\sin\theta = \frac{\sqrt{7}}{4}$, $\tan\theta = \boxed{\frac{\sqrt{7}}{3}}$

(c) $V = \pi\int_{\theta}^{\pi/4} \sec^2 x\,dx = \pi\left[\tan x\right]_{\theta}^{\pi/4} = \boxed{\pi\left(1 - \frac{\sqrt{7}}{3}\right)}$

11

May 2017 · Paper 2 · TZ2 · Q10

The following table shows a probability distribution for the random variable X, where $\mathrm{E}(X) = 1.2$.

x	0	1	2	3
$\mathrm{P}(X = x)$	p	$\frac{1}{2}$	$\frac{3}{10}$	q

(a) (i) Find q.

(ii) Find p. [4]

A bag contains white and blue marbles, with at least three of each colour. Three marbles are drawn from the bag, without replacement. The number of blue marbles drawn is given by the random variable X.

(b) (i) Write down the probability of drawing three blue marbles.

(ii) Explain why the probability of drawing three white marbles is $\frac{1}{6}$.

(iii) The bag contains a total of ten marbles of which w are white. Find w. [5]

A game is played in which three marbles are drawn from the bag of ten marbles, without replacement. A player wins a prize if three white marbles are drawn.

(c) Jill plays the game nine times. Find the probability that she wins exactly two prizes. [2]

(d) Grant plays the game until he wins two prizes. Find the probability that he wins his second prize on his eighth attempt. [4]

11 *May 2017 · Paper 2 · TZ2 · Q10*

(a)(i) $\frac{1}{2}+\frac{6}{10}+3q=1.2 \Rightarrow \boxed{q=\frac{1}{30}}$ (ii) $p=1-\frac{1}{2}-\frac{3}{10}-\frac{1}{30}=\boxed{\frac{1}{6}}$

(b)(i) $\boxed{\frac{1}{30}}$ (ii) three white $\Leftrightarrow X=0$, so $p=\frac{1}{6}$ (iii) $\frac{w(w-1)(w-2)}{10\cdot 9\cdot 8}=\frac{1}{6} \Rightarrow$ $\boxed{w=6}$

(c) $Y \sim \mathrm{B}(9,\frac{1}{6})$: $\mathrm{P}(Y=2)=\boxed{0.279}$ (d) $7\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^6\times\frac{1}{6}=\boxed{0.0651}$

12

May 2016 · Paper 2 · TZ1 · Q8

A factory has two machines, A and B. The number of breakdowns of each machine is independent from day to day.

Let A be the number of breakdowns of Machine A on any given day. The probability distribution for A can be modelled by the following table.

a	0	1	2	3
$\mathrm{P}(A = a)$	0.55	0.3	0.1	k

(a) Find k. [2]

(b) (i) A day is chosen at random. Write down the probability that Machine A has no breakdowns.

(ii) Five days are chosen at random. Find the probability that Machine A has no breakdowns on exactly four of these days. [3]

Let B be the number of breakdowns of Machine B on any given day. The probability distribution for B can be modelled by the following table.

b	0	1	2	3
$\mathrm{P}(B = b)$	0.7	0.2	0.08	0.02

(c) Find $\mathrm{E}(B)$. [2]

On Tuesday, the factory uses both Machine A and Machine B. The variables A and B are independent.

(d) (i) Find the probability that there are exactly two breakdowns on Tuesday.

(ii) Given that there are exactly two breakdowns on Tuesday, find the probability that both breakdowns are of Machine A. [8]

12 *May 2016 · Paper 2 · TZ1 · Q8*

(a) $\boxed{k = 0.05}$ (b)(i) $\boxed{0.55}$ (ii) $\binom{5}{4}(0.55)^4(0.45) = \boxed{0.206}$ (c) $0.2 + 0.16 + 0.06 = \boxed{0.42}$

(d)(i) $0.55(0.08) + 0.3(0.2) + 0.1(0.7) = \boxed{0.174}$ (ii) $\frac{0.07}{0.174} = \boxed{0.402}$

13

November 2015 · Paper 2 · Q2

The following table shows the probability distribution of a discrete random variable X.

x	0	1	2	3
$\mathrm{P}(X=x)$	0.15	k	0.1	$2k$

(a) Find the value of k. [3]

(b) Find $\mathrm{E}(X)$. [2]

13 *November 2015 · Paper 2 · Q2*

(a) $0.25 + 3k = 1 \Rightarrow \boxed{k = 0.25}$ (b) $0.25 + 0.2 + 1.5 = \boxed{1.95}$

14

May 2015 · Paper 1 · TZ1 · Q1

A discrete random variable X has the following probability distribution.

x	0	1	2	3
$\mathrm{P}(X=x)$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{2}{10}$	p

(a) Find p. [3]

(b) Find $\mathrm{E}(X)$. [3]

14 *May 2015 · Paper 1 · TZ1 · Q1*

(a) $p = 1 - \frac{9}{10} = \boxed{\frac{1}{10}}$ (b) $\frac{0+4+4+3}{10} = \boxed{\frac{11}{10}}$

15

May 2015 · Paper 1 · TZ2 · Q7

[Maximum mark: 7]

A bag contains black and white chips. Rose pays $\$10$ to play a game where she draws a chip from the bag. The following table gives the probability of choosing each colour chip.

Outcome	black	white
Probability	0.4	0.6

Rose gets no money if she draws a white chip, and gets $\$k$ if she draws a black chip. The game is fair. Find the value of k.

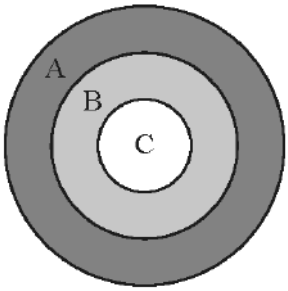
15 *May 2015 · Paper 1 · TZ2 · Q7*

Fair game: expected winnings = cost. $0.4k + 0.6(0) = 10 \Rightarrow \boxed{k = 25}$

16

November 2014 · Paper 1 · Q3

The following diagram shows a board which is divided into three regions A, B and C.



A game consists of a contestant throwing one dart at the board. The probability of hitting each region is given in the following table.

Region	A	B	C
Probability	$\frac{5}{20}$	$\frac{4}{20}$	$\frac{1}{20}$

(a) Find the probability that the dart does **not** hit the board. *[3]*

The contestant scores points as shown in the following table.

Region	A	B	C	Does not hit the board
Points	0	q	10	−3

(b) Given that the game is fair, find the value of q. *[4]*

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16 *November 2014 · Paper 1 · Q3*

(a) $1 - \frac{5+4+1}{20} = \boxed{\frac{1}{2}}$

(b) Fair $\Rightarrow$ E $= 0$: $0 + \frac{4q}{20} + 10\left(\frac{1}{20}\right) - 3\left(\frac{1}{2}\right) = 0 \Rightarrow \frac{q}{5} = 1 \Rightarrow \boxed{q = 5}$

17

November 2012 · Paper 1 · Q2

The following table shows the probability distribution of a discrete random variable X.

x	0	2	5	9
$P(X = x)$	0.3	k	$2k$	0.1

(a) Find the value of k. *[3 marks]*

(b) Find $E(X)$. *[3 marks]*

17 *November 2012 · Paper 1 · Q2*

(a) $0.3+3k+0.1=1 \Rightarrow \boxed{k=0.2}$ (b) $2(0.2)+5(0.4)+9(0.1)=\boxed{3.3}$

18

May 2012 · Paper 1 · TZ2 · Q4

The random variable X has the following probability distribution, with $P(X > 1) = 0.5$.

x	0	1	2	3
$P(X = x)$	p	q	r	0.2

(a) Find the value of r. *[2 marks]*

(b) Given that $E(X) = 1.4$, find the value of p and of q. *[6 marks]*

18 *May 2012 · Paper 1 · TZ2 · Q4*

(a) $\mathrm{P}(X > 1) = r + 0.2 = 0.5 \Rightarrow \boxed{r = 0.3}$

(b) $p + q = 0.5$ and $q + 2(0.3) + 3(0.2) = 1.4 \Rightarrow \boxed{q = 0.2,\ p = 0.3}$

19

November 2021 · Paper 2 · Q4

[Maximum mark: 7]

A factory manufactures lamps. It is known that the probability that a lamp is found to be defective is 0.05. A random sample of 30 lamps is tested.

(a) Find the probability that there is at least one defective lamp in the sample. [3]

(b) Given that there is at least one defective lamp in the sample, find the probability that there are at most two defective lamps. [4]

20

May 2017 · Paper 2 · TZ1 · Q4

In a large university the probability that a student is left handed is 0.08. A sample of 150 students is randomly selected from the university. Let k be the expected number of left-handed students in this sample.

(a) Find k. [2]

(b) Hence, find the probability that

(i) exactly k students are left handed;

(ii) fewer than k students are left handed. [4]

19 *November 2021 · Paper 2 · Q4*

$X \sim \mathrm{B}(30, 0.05)$. (a) $1 - 0.95^{30} = \boxed{0.785}$ (b) $\dfrac{\mathrm{P}(X=1)+\mathrm{P}(X=2)}{\mathrm{P}(X \geq 1)} = \dfrac{0.5975}{0.7854} = \boxed{0.761}$

20 *May 2017 · Paper 2 · TZ1 · Q4*

(a) $k = 150 \times 0.08 = \boxed{12}$ (b)(i) $\mathrm{P}(X = 12) = \boxed{0.119}$ (ii) $\mathrm{P}(X \leq 11) = \boxed{0.457}$

21

November 2016 · Paper 2 · Q7

A jar contains 5 red discs, 10 blue discs and m green discs. A disc is selected at random and replaced. This process is performed four times.

(a) Write down the probability that the first disc selected is red. [1]

(b) Let X be the number of red discs selected. Find the smallest value of m for which $\text{Var}(X) < 0.6$. [5]

22

November 2013 · Paper 2 · Q10

Samantha goes to school five days a week. When it rains, the probability that she goes to school by bus is 0.5. When it does not rain, the probability that she goes to school by bus is 0.3. The probability that it rains on any given day is 0.2.

(a) On a randomly selected school day, find the probability that Samantha goes to school by bus. *[4]*

(b) Given that Samantha went to school by bus on Monday, find the probability that it was raining. *[3]*

(c) In a randomly chosen school week, find the probability that Samantha goes to school by bus on exactly three days. *[2]*

(d) After n school days, the probability that Samantha goes to school by bus at least once is greater than 0.95. Find the smallest value of n. *[5]*

21 *November 2016 · Paper 2 · Q7*

(a) $\boxed{\frac{5}{15+m}}$

(b) $X \sim \mathrm{B}(4,p)$, $\mathrm{Var} = 4p(1-p) < 0.6 \Rightarrow p < 0.1838$. $\frac{5}{15+m} < 0.1838 \Rightarrow m > 12.2 \Rightarrow \boxed{m = 13}$

22 *November 2013 · Paper 2 · Q10*

(a) $0.2(0.5)+0.8(0.3) = \boxed{0.34}$ (b) $\frac{0.1}{0.34} = \boxed{0.294}$

(c) $\mathrm{B}(5,0.34)$: $\mathrm{P}(X=3) = \boxed{0.171}$ (d) $1-0.66^n > 0.95 \Rightarrow n > 7.21 \Rightarrow \boxed{n=8}$

23

May 2013 · Paper 2 · TZ2 · Q9

A bag contains four gold balls and six silver balls.

(a) Two balls are drawn at random from the bag, with replacement. Let X be the number of gold balls drawn from the bag.

(i) Find $\mathrm{P}(X = 0)$.

(ii) Find $\mathrm{P}(X = 1)$.

(iii) Hence, find $\mathrm{E}(X)$. *[8 marks]*

Fourteen balls are drawn from the bag, with replacement.

(b) Find the probability that exactly five of the balls are gold. *[2 marks]*

(c) Find the probability that at most five of the balls are gold. *[2 marks]*

(d) Given that at most five of the balls are gold, find the probability that exactly five of the balls are gold. Give the answer correct to two decimal places. *[3 marks]*

23 *May 2013 · Paper 2 · TZ2 · Q9*

(a)(i) $0.6^2 = \boxed{0.36}$ (ii) $2(0.4)(0.6) = \boxed{0.48}$ (iii) $0.48 + 2(0.16) = \boxed{0.8}$

(b) $\mathrm{B}(14, 0.4)$: $\mathrm{P}(X = 5) = \boxed{0.207}$ (c) $\mathrm{P}(X \leq 5) = \boxed{0.486}$ (d) $\frac{0.2066}{0.4859} = \boxed{0.43}$

24

May 2012 · Paper 2 · TZ1 · Q7

The probability of obtaining "tails" when a biased coin is tossed is 0.57. The coin is tossed ten times. Find the probability of obtaining

(a) **at least** four tails; *[4 marks]*

(b) the fourth tail on the tenth toss. *[3 marks]*

24 *May 2012 · Paper 2 · TZ1 · Q7*

$X \sim \mathrm{B}(10, 0.57)$. (a) $\mathrm{P}(X \geq 4) = \boxed{0.919}$

(b) 3 tails in the first 9, then a tail: $\binom{9}{3}(0.57)^3(0.43)^6 \times 0.57 = \boxed{0.0561}$

25

November 2021 · Paper 2 · Q9

[Maximum mark: 16]

The random variable X follows a normal distribution with mean μ and standard deviation σ.

(a) Find $\mathrm{P}(\mu - 1.5\sigma < X < \mu + 1.5\sigma)$. [3]

The avocados grown on a farm have weights, in grams, that are normally distributed with mean μ and standard deviation σ. Avocados are categorized as small, medium, large or premium, according to their weight. The following table shows the probability an avocado grown on the farm is classified as small, medium, large or premium.

Category	Small	Medium	Large	Premium
Probability	0.04	0.576	0.288	0.096

The maximum weight of a small avocado is 106.2 grams.

The minimum weight of a premium avocado is 182.6 grams.

(b) Find the value of μ and of σ. [5]

A supermarket purchases all the avocados from the farm that weigh more than 106.2 grams.

(c) Find the probability that an avocado chosen at random from this purchase is categorized as

(i) medium;

(ii) large;

(iii) premium. [4]

The selling prices of the different categories of avocado at this supermarket are shown in the following table:

Category	Medium	Large	Premium
Selling price ($) per avocado	1.10	1.29	1.96

The supermarket pays the farm $\$200$ for the avocados and assumes it will then sell them in exactly the same proportion as purchased from the farm.

(d) According to this model, find the minimum number of avocados that must be sold so that the net profit for the supermarket is at least $\$438$. [4]

25 *November 2021 · Paper 2 · Q9*

(a) $\mathrm{P}(-1.5 < Z < 1.5) = \boxed{0.866}$

(b) $\frac{106.2-\mu}{\sigma} = -1.7507,\ \frac{182.6-\mu}{\sigma} = 1.3047 \Rightarrow \boxed{\mu = 150,\ \sigma = 25}$

(c) divide by 0.96: (i) $\boxed{0.6}$ (ii) $\boxed{0.3}$ (iii) $\boxed{0.1}$

(d) mean income per avocado $= 1.10(0.6) + 1.29(0.3) + 1.96(0.1) = 1.243;\ 1.243n - 200 \geq 438 \Rightarrow n \geq 513.3$: $\boxed{514}$

26

May 2021 · Paper 2 · TZ1 · Q3

[Maximum mark: 6]

A company produces bags of sugar whose masses, in grams, can be modelled by a normal distribution with mean 1000 and standard deviation 3.5. A bag of sugar is rejected for sale if its mass is less than 995 grams.

(a) Find the probability that a bag selected at random is rejected. [2]

(b) Estimate the number of bags which will be rejected from a random sample of 100 bags. [1]

(c) Given that a bag is not rejected, find the probability that it has a mass greater than 1005 grams. [3]

26 *May 2021 · Paper 2 · TZ1 · Q3*

(a) $\mathrm{P}(X < 995) = \boxed{0.0766}$ (b) $100 \times 0.0766 = \boxed{7.66\ (\approx 8)}$

(c) $\dfrac{\mathrm{P}(X > 1005)}{\mathrm{P}(X \geq 995)} = \dfrac{0.0766}{0.9234} = \boxed{0.0829}$

27

November 2020 · Paper 2 · Q9

Fiona walks from her house to a bus stop where she gets a bus to school. Her time, W minutes, to walk to the bus stop is normally distributed with $W \sim \mathrm{N}(12, 3^2)$.

Fiona always leaves her house at $07{:}15$. The first bus that she can get departs at $07{:}30$.

(a) Find the probability that it will take Fiona between 15 minutes and 30 minutes to walk to the bus stop. [2]

The length of time, B minutes, of the bus journey to Fiona's school is normally distributed with $B \sim \mathrm{N}(50, \sigma^2)$. The probability that the bus journey takes less than 60 minutes is 0.941.

(b) Find σ. [3]

(c) Find the probability that the bus journey takes less than 45 minutes. [2]

If Fiona misses the first bus, there is a second bus which departs at $07{:}45$. She must arrive at school by $08{:}30$ to be on time. Fiona will not arrive on time if she misses both buses. The variables W and B are independent.

(d) Find the probability that Fiona will arrive on time. [5]

This year, Fiona will go to school on 183 days.

(e) Calculate the number of days Fiona is expected to arrive on time. [2]

27 *November 2020 · Paper 2 · Q9*

(a) $\boxed{0.159}$ (b) $\frac{60-50}{\sigma} = 1.5632 \Rightarrow \boxed{\sigma = 6.40}$ (c) $\boxed{0.217}$

(d) $\mathrm{P}(W < 15) \times 0.941 + \mathrm{P}(15 < W < 30) \times \mathrm{P}(B < 45) = 0.841(0.941) + 0.159(0.217) = \boxed{0.826}$

(e) $183 \times 0.826 = \boxed{151 \text{ days}}$

28

November 2019 · Paper 1 · Q7

[Maximum mark: 6]

Let X and Y be normally distributed with $X \sim \mathrm{N}(14, a^2)$ and $Y \sim \mathrm{N}(22, a^2)$, $a > 0$.

(a) Find b so that $\mathrm{P}(X > b) = \mathrm{P}(Y < b)$. [2]

It is given that $\mathrm{P}(X > 20) = 0.112$.

(b) Find $\mathrm{P}(16 < Y < 28)$. [4]

28 *November 2019 · Paper 1 · Q7*

(a) Same spread, symmetry: b is halfway between the means, $\boxed{b = 18}$

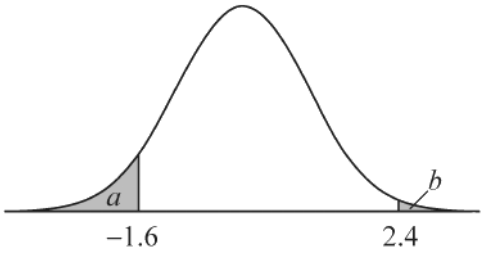
(b) $\mathrm{P}(X > 20) = \mathrm{P}(Z > \frac{6}{a}) = 0.112$, so $\mathrm{P}(Y < 16) = \mathrm{P}(Y > 28) = 0.112$.

$\mathrm{P}(16 < Y < 28) = 1 - 2(0.112) = \boxed{0.776}$

29

May 2019 · Paper 1 · TZ1 · Q9

A random variable Z is normally distributed with mean 0 and standard deviation 1. It is known that $\mathrm{P}(z < -1.6) = a$ and $\mathrm{P}(z > 2.4) = b$. This is shown in the following diagram.



(a) Find $\mathrm{P}(-1.6 < z < 2.4)$. Write your answer in terms of a and b. [2]

(b) Given that $z > -1.6$, find the probability that $z < 2.4$. Write your answer in terms of a and b. [4]

A second random variable X is normally distributed with mean m and standard deviation s.

It is known that $\mathrm{P}(x < 1) = a$.

(c) Write down the standardized value for $x = 1$. [1]

It is also known that $\mathrm{P}(x > 2) = b$.

(d) Find s. [6]

29 *May 2019 · Paper 1 · TZ1 · Q9*

(a) $\boxed{1-a-b}$ (b) $\boxed{\dfrac{1-a-b}{1-a}}$ (c) $\boxed{z=-1.6}$

(d) $\frac{1-m}{s}=-1.6$, $\frac{2-m}{s}=2.4$. Subtract: $\frac{1}{s}=4 \Rightarrow \boxed{s=0.25}$

30

May 2019 · Paper 2 · TZ2 · Q9

At Penna Airport the probability, $\mathrm{P}(A)$, that all passengers arrive on time for a flight is 0.70. The probability, $\mathrm{P}(D)$, that a flight departs on time is 0.85. The probability that all passengers arrive on time for a flight and it departs on time is 0.65.

(a) Show that event A and event D are **not** independent. [2]

(b) (i) Find $\mathrm{P}(A \cap D')$.

(ii) Given that all passengers for a flight arrive on time, find the probability that the flight does **not** depart on time. [5]

The number of hours that pilots fly per week is normally distributed with a mean of 25 hours and a standard deviation σ. $90\,\%$ of pilots fly less than 28 hours in a week.

(c) Find the value of σ. [3]

(d) All flights have two pilots. Find the percentage of flights where **both** pilots flew more than 30 hours last week. [4]

31

November 2017 · Paper 2 · Q7

[Maximum mark: 7]

The heights of adult males in a country are normally distributed with a mean of $180\,\mathrm{cm}$ and a standard deviation of $\sigma\,\mathrm{cm}$. $17\,\%$ of these men are shorter than $168\,\mathrm{cm}$. $80\,\%$ of them have heights between $(192 - h)\,\mathrm{cm}$ and $192\,\mathrm{cm}$.

Find the value of h.

30 *May 2019 · Paper 2 · TZ2 · Q9*

(a) $P(A)P(D) = 0.7 \times 0.85 = 0.595 \neq 0.65 = P(A \cap D)$ ✓

(b)(i) $0.7 - 0.65 = \boxed{0.05}$ (ii) $\frac{0.05}{0.7} = \boxed{0.0714}$ (c) $\frac{3}{\sigma} = 1.2816 \Rightarrow \boxed{\sigma = 2.34}$

(d) $P(X > 30) = 0.01635$; both: $0.01635^2 = 0.000267 = \boxed{0.0267\%}$

31 *November 2017 · Paper 2 · Q7*

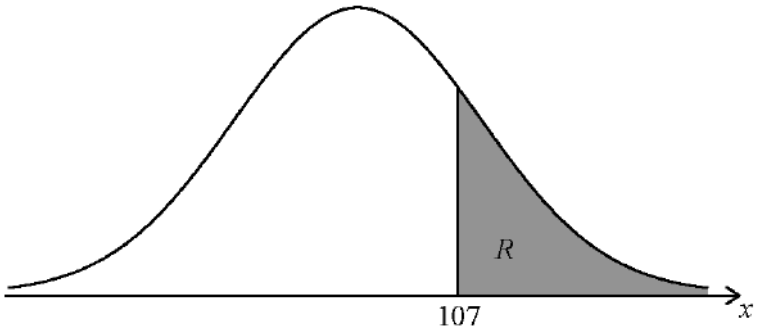
$\frac{168-180}{\sigma} = -0.9542 \Rightarrow \sigma = 12.58.$ $P(X < 192) = 0.8296$, so $P(X < 192 - h) = 0.0296$

$192 - h = 180 + 12.58(-1.887) = 156.3 \Rightarrow \boxed{h = 35.7}$

32

May 2017 · Paper 1 · TZ2 · Q3

The random variable X is normally distributed with a mean of 100. The following diagram shows the normal curve for X.



Let R be the shaded region under the curve, to the right of 107. The area of R is 0.24.

(a) Write down $\mathrm{P}(X > 107)$. [1]

(b) Find $\mathrm{P}(100 < X < 107)$. [3]

(c) Find $\mathrm{P}(93 < X < 107)$. [2]

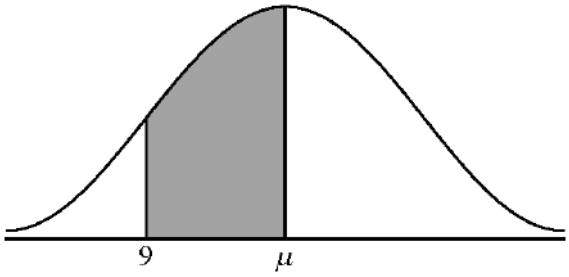
32 *May 2017 · Paper 1 · TZ2 · Q3*

(a) $\boxed{0.24}$ (b) $0.5 - 0.24 = \boxed{0.26}$ (c) symmetry: $2(0.26) = \boxed{0.52}$

33

May 2017 · Paper 2 · TZ1 · Q9

A random variable X is normally distributed with mean, μ. In the following diagram, the shaded region between 9 and μ represents $30\,\%$ of the distribution.



(a) Find $P(X < 9)$. [2]

The standard deviation of X is 2.1.

(b) Find the value of μ. [3]

The random variable Y is normally distributed with mean λ and standard deviation 3.5.
The events $X > 9$ and $Y > 9$ are independent, and $P\big((X > 9) \cap (Y > 9)\big) = 0.4$.

(c) Find λ. [5]

(d) Given that $Y > 9$, find $P(Y < 13)$. [5]

33 *May 2017 · Paper 2 · TZ1 · Q9*

(a) $0.5 - 0.3 = \boxed{0.2}$ (b) $\frac{9-\mu}{2.1} = -0.8416 \Rightarrow \boxed{\mu = 10.8}$

(c) $0.8 \times \mathrm{P}(Y > 9) = 0.4 \Rightarrow \mathrm{P}(Y > 9) = 0.5 \Rightarrow \boxed{\lambda = 9}$

(d) $\dfrac{\mathrm{P}(9 < Y < 13)}{0.5} = \dfrac{0.37345}{0.5} = \boxed{0.747}$

34

November 2016 · Paper 2 · Q5

The weights, W, of newborn babies in Australia are normally distributed with a mean $3.41\,\text{kg}$ and standard deviation $0.57\,\text{kg}$. A newborn baby has a low birth weight if it weighs less than $w\,\text{kg}$.

(a) Given that 5.3% of newborn babies have a low birth weight, find w. [3]

(b) A newborn baby has a low birth weight.
Find the probability that the baby weighs at least $2.15\,\text{kg}$. [3]

34 *November 2016 · Paper 2 · Q5*

(a) $\frac{w-3.41}{0.57} = -1.6164 \Rightarrow \boxed{w = 2.49}$ (b) $\frac{\mathrm{P}(2.15 \leq W < 2.489)}{0.053} = \boxed{0.745}$

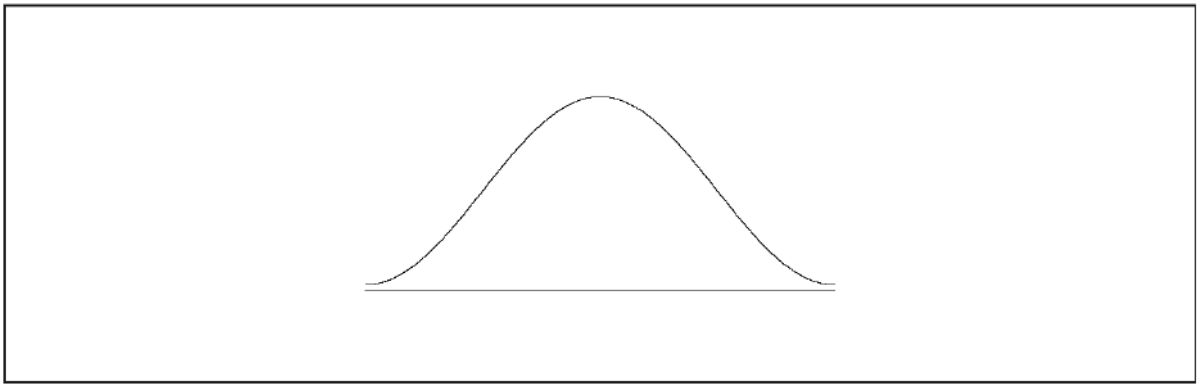
35

May 2016 · Paper 2 · TZ1 · Q1

[Maximum mark: 6]

A random variable X is distributed normally with a mean of 20 and standard deviation of 4.

(a) On the following diagram, shade the region representing $\mathrm{P}(X \leq 25)$. [2]



(b) Write down $\mathrm{P}(X \leq 25)$, correct to two decimal places. [2]

(c) Let $\mathrm{P}(X \leq c) = 0.7$. Write down the value of c. [2]

35 *May 2016 · Paper 2 · TZ1 · Q1*

(a) Shade the region to the **left** of $x = 25$ (25 is just right of the mean 20). (b) $\boxed{0.89}$ (c) $c = 20 + 4(0.5244) = \boxed{22.1}$

36

May 2016 · Paper 2 · TZ2 · Q6

A competition consists of two independent events, shooting at 100 targets and running for one hour.

The number of targets a contestant hits is the S score. The S scores are normally distributed with mean 65 and standard deviation 10.

(a) A contestant is chosen at random. Find the probability that their S score is less than 50. [2]

The distance in km that a contestant runs in one hour is the R score. The R scores are normally distributed with mean 12 and standard deviation 2.5. The R score is independent of the S score.

Contestants are disqualified if their S score is less than 50 **and** their R score is less than $x\,\text{km}$.

(b) Given that $1\,\%$ of the contestants are disqualified, find the value of x. [4]

36 *May 2016 · Paper 2 · TZ2 · Q6*

(a) $\mathrm{P}(S < 50) = \boxed{0.0668}$ (b) $0.0668 \times \mathrm{P}(R < x) = 0.01 \Rightarrow \mathrm{P}(R < x) = 0.1497 \Rightarrow \boxed{x = 9.41}$

37

November 2015 · Paper 2 · Q10

The masses of watermelons grown on a farm are normally distributed with a mean of $10\,\text{kg}$. The watermelons are classified as small, medium or large.

A watermelon is small if its mass is less than $4\,\text{kg}$. Five percent of the watermelons are classified as small.

(a) Find the standard deviation of the masses of the watermelons. [4]

The following table shows the percentages of small, medium and large watermelons grown on the farm.

small	medium	large
5%	57%	38%

A watermelon is large if its mass is greater than $w\,\text{kg}$.

(b) Find the value of w. [2]

All the medium and large watermelons are delivered to a grocer.

(c) The grocer selects a watermelon at random from **this** delivery. Find the probability that it is medium. [3]

(d) The grocer sells all the medium watermelons for $\$1.75$ each, and all the large watermelons for $\$3.00$ each. His costs on this delivery are $\$300$, and his total profit is $\$150$. Find the number of watermelons in the delivery. [5]

37 *November 2015 · Paper 2 · Q10*

(a) $\frac{4-10}{\sigma} = -1.6449 \Rightarrow \boxed{\sigma = 3.65}$ (b) $\mathrm{P}(M < w) = 0.62 \Rightarrow \boxed{w = 11.1}$

(c) $\frac{0.57}{0.95} = \boxed{0.6}$ (d) $\text{income} = 0.6n(1.75) + 0.4n(3) = 2.25n;\ 2.25n - 300 = 150 \Rightarrow \boxed{n = 200}$

38

November 2014 · Paper 2 · Q10

The weights of fish in a lake are normally distributed with a mean of 760 g and standard deviation σ. It is known that 78.87% of the fish have weights between 705 g and 815 g.

(a) (i) Write down the probability that a fish weighs more than 760 g.

(ii) Find the probability that a fish weighs less than 815 g. *[4]*

(b) (i) Write down the standardized value for 815 g.

(ii) Hence or otherwise, find σ. *[4]*

A fishing contest takes place in the lake. Small fish, called tiddlers, are thrown back into the lake. The maximum weight of a tiddler is 1.5 standard deviations below the mean.

(c) Find the maximum weight of a tiddler. *[2]*

(d) A fish is caught at random. Find the probability that it is a tiddler. *[2]*

(e) 25% of the fish in the lake are salmon. 10% of the salmon are tiddlers. Given that a fish caught at random is a tiddler, find the probability that it is a salmon. *[4]*

38 *November 2014 · Paper 2 · Q10*

(a)(i) $\boxed{0.5}$ (ii) symmetric about 760: $0.5 + \frac{0.7887}{2} = \boxed{0.894}$

(b)(i) $\boxed{z = 1.25}$ (ii) $\frac{815-760}{\sigma} = 1.25 \Rightarrow \boxed{\sigma = 44}$

(c) $760 - 1.5(44) = \boxed{694 \text{ g}}$ (d) $\mathrm{P}(Z < -1.5) = \boxed{0.0668}$ (e) $\dfrac{0.25 \times 0.1}{0.0668} = \boxed{0.374}$

39

November 2013 · Paper 2 · Q6

The time taken for a student to complete a task is normally distributed with a mean of 20 minutes and a standard deviation of 1.25 minutes.

(a) A student is selected at random. Find the probability that the student completes the task in less than 21.8 minutes. *[2]*

(b) The probability that a student takes between k and 21.8 minutes is 0.3. Find the value of k. *[5]*

39 *November 2013 · Paper 2 · Q6*

(a) $\mathrm{P}(T < 21.8) = \boxed{0.925}$ (b) $\mathrm{P}(T < k) = 0.925 - 0.3 = 0.625 \Rightarrow k =$

$20 + 1.25(0.3186) = \boxed{20.4}$

40

May 2013 · Paper 2 · TZ2 · Q2

The random variable X is normally distributed with mean 20 and standard deviation 5.

(a) Find $P(X \leq 22.9)$. *[3 marks]*

(b) Given that $P(X < k) = 0.55$, find the value of k. *[3 marks]*

40 *May 2013 · Paper 2 · TZ2 · Q2*

(a) $\boxed{0.719}$ (b) $k = 20 + 5(0.1257) = \boxed{20.6}$

41

May 2013 · Paper 2 · TZ1 · Q7

[Maximum mark: 7]

A random variable X is normally distributed with $\mu = 150$ and $\sigma = 10$.

Find the interquartile range of X.

41 *May 2013 · Paper 2 · TZ1 · Q7*

$Q_1 = 150 - 10(0.6745)$, $Q_3 = 150 + 10(0.6745)$; $\text{IQR} = 20(0.6745) =$

$\boxed{13.5}$

42

November 2012 · Paper 2 · Q6

[Maximum mark: 8]

In a large city, the time taken to travel to work is normally distributed with mean μ and standard deviation σ. It is found that 4 % of the population take less than 5 minutes to get to work, and 70 % take less than 25 minutes.

Find the value of μ and of σ.

42 *November 2012 · Paper 2 · Q6*

$\frac{5-\mu}{\sigma} = -1.7507$, $\frac{25-\mu}{\sigma} = 0.5244$. Subtract: $20 = 2.2751\sigma \Rightarrow \boxed{\sigma = 8.79}$, $\boxed{\mu = 20.4}$

43

May 2012 · Paper 2 · TZ2 · Q4

The heights of a group of seven-year-old children are normally distributed with mean 117 cm and standard deviation 5 cm. A child is chosen at random from the group.

(a) Find the probability that this child is taller than 122.5 cm. *[3 marks]*

(b) The probability that this child is shorter than k cm is 0.65. Find the value of k. *[3 marks]*

43 *May 2012 · Paper 2 · TZ2 · Q4*

(a) $\mathrm{P}(X > 122.5) = \boxed{0.136}$ (b) $k = 117 + 5(0.3853) = \boxed{118.9}$

44

May 2021 · Paper 2 · TZ2 · Q8

[Maximum mark: 15]

The flight times, T minutes, between two cities can be modelled by a normal distribution with a mean of 75 minutes and a standard deviation of σ minutes.

(a) Given that 2% of the flight times are longer than 82 minutes, find the value of σ. [3]

(b) Find the probability that a randomly selected flight will have a flight time of more than 80 minutes. [2]

(c) Given that a flight between the two cities takes longer than 80 minutes, find the probability that it takes less than 82 minutes. [4]

On a particular day, there are 64 flights scheduled between these two cities.

(d) Find the expected number of flights that will have a flight time of more than 80 minutes. [3]

(e) Find the probability that more than 6 of the flights on this particular day will have a flight time of more than 80 minutes. [3]

44 *May 2021 · Paper 2 · TZ2 · Q8*

(a) $\frac{82-75}{\sigma} = 2.0537 \Rightarrow \boxed{\sigma = 3.41}$ (b) $\mathrm{P}(T > 80) = \boxed{0.0712}$

(c) $\dfrac{\mathrm{P}(80 < T < 82)}{\mathrm{P}(T > 80)} = \dfrac{0.0512}{0.0712} = \boxed{0.719}$ (d) $64 \times 0.0712 = \boxed{4.56}$

(e) $Y \sim \mathrm{B}(64, 0.0712)$: $\mathrm{P}(Y > 6) = 1 - \mathrm{P}(Y \leq 6) = \boxed{0.169}$

45

November 2019 · Paper 2 · Q9

[Maximum mark: 15]

SpeedWay airline flies from city A to city B. The flight time is normally distributed with a mean of 260 minutes and a standard deviation of 15 minutes.

A flight is considered late if it takes longer than 275 minutes.

(a) Calculate the probability a flight is **not** late. [2]

The flight is considered to be **on time** if it takes between m and 275 minutes. The probability that a flight is on time is 0.830.

(b) Find the value of m. [3]

During a week, SpeedWay has 12 flights from city A to city B. The time taken for any flight is independent of the time taken by any other flight.

(c) (i) Calculate the probability that at least 7 of these flights are **on time**.

(ii) Given that at least 7 of these flights are on time, find the probability that exactly 10 flights are on time. [7]

SpeedWay increases the number of flights from city A to city B to 20 flights each week, and improves their efficiency so that more flights are on time. The probability that at least 19 flights are on time is 0.788.

(d) A flight is chosen at random. Calculate the probability that it is on time. [3]

45 *November 2019 · Paper 2 · Q9*

(a) $\mathrm{P}(T < 275) = \boxed{0.841}$ (b) $\mathrm{P}(T < m) = 0.8413 - 0.830 = 0.0113 \Rightarrow$ $\boxed{m = 226}$

(c)(i) $\mathrm{B}(12, 0.83)$: $\mathrm{P}(X \geq 7) = \boxed{0.991}$ (ii) $\dfrac{\mathrm{P}(X = 10)}{\mathrm{P}(X \geq 7)} = \boxed{0.299}$

(d) $p^{20} + 20p^{19}(1-p) = 0.788 \Rightarrow \boxed{p = 0.957}$

46

November 2018 · Paper 2 · Q9

A nationwide study on reaction time is conducted on participants in two age groups. The participants in Group X are less than 40 years old. Their reaction times are normally distributed with mean 0.489 seconds and standard deviation 0.07 seconds.

(a) A person is selected at random from Group X. Find the probability that their reaction time is greater than 0.65 seconds. [2]

The participants in Group Y are 40 years or older. Their reaction times are normally distributed with mean 0.592 seconds and standard deviation σ seconds.

(b) The probability that the reaction time of a person in Group Y is greater than 0.65 seconds is 0.396. Find the value of σ. [4]

In the study, 38% of the participants are in Group X.

(c) A randomly selected participant has a reaction time greater than 0.65 seconds. Find the probability that the participant is in Group X. [6]

(d) Ten of the participants with reaction times greater than 0.65 are selected at random. Find the probability that at least two of them are in Group X. [3]

46 *November 2018 · Paper 2 · Q9*

(a) $\mathrm{P}(X > 0.65) = \boxed{0.0107}$ (b) $\frac{0.65-0.592}{\sigma} = 0.2637 \Rightarrow \boxed{\sigma = 0.220}$

(c) $\dfrac{0.38(0.01072)}{0.38(0.01072)+0.62(0.396)} = \boxed{0.0163}$ (d) $\mathrm{B}(10, 0.0163)$: $\mathrm{P}(\geq 2) = \boxed{0.0110}$

47

May 2018 · Paper 2 · TZ2 · Q10

The mass M of apples in grams is normally distributed with mean μ. The following table shows probabilities for values of M.

Values of M	$M < 93$	$93 \leq M \leq 119$	$M > 119$
$P(X)$	k	0.98	0.01

(a) (i) Write down the value of k.

(ii) Show that $\mu = 106$. [4]

(b) Find $P(M < 95)$. [5]

The apples are packed in bags of ten.

Any apples with a mass less than $95\,\text{g}$ are classified as small.

(c) Find the probability that a bag of apples selected at random contains at most one small apple. [3]

(d) A crate contains 50 bags of apples. A crate is selected at random.

(i) Find the expected number of bags in this crate that contain at most one small apple.

(ii) Find the probability that at least 48 bags in this crate contain at most one small apple. [5]

47 *May 2018 · Paper 2 · TZ2 · Q10*

(a)(i) $\boxed{k = 0.01}$ (ii) $\mathrm{P}(M < 93) = \mathrm{P}(M > 119)$, so $\mu = \frac{93+119}{2} = 106$ ✓

(b) $\sigma = \frac{13}{2.3263} = 5.588$; $\mathrm{P}(M < 95) = \boxed{0.0245}$

(c) $\mathrm{B}(10, 0.0245)$: $\mathrm{P}(\leq 1) = \boxed{0.976}$ (d)(i) $50(0.976) = \boxed{48.8}$ (ii) $\mathrm{B}(50, 0.976)$: $\mathrm{P}(\geq 48) = \boxed{0.885}$

48

May 2018 · Paper 2 · TZ1 · Q9

The weights, in grams, of oranges grown in an orchard, are normally distributed with a mean of $297\,\text{g}$. It is known that $79\,\%$ of the oranges weigh more than $289\,\text{g}$ and $9.5\,\%$ of the oranges weigh more than $310\,\text{g}$.

(a) Find the probability that an orange weighs between $289\,\text{g}$ and $310\,\text{g}$. [2]

The weights of the oranges have a standard deviation of σ.

(b) (i) Find the standardized value for $289\,\text{g}$.

(ii) Hence, find the value of σ. [5]

The grocer at a local grocery store will buy the oranges whose weights exceed the 35th percentile.

(c) To the nearest gram, find the minimum weight of an orange that the grocer will buy. [3]

The orchard packs oranges in boxes of 36.

(d) Find the probability that the grocer buys more than half the oranges in a box selected at random. [5]

The grocer selects two boxes at random.

(e) Find the probability that the grocer buys more than half the oranges in each box. [2]

48 *May 2018 · Paper 2 · TZ1 · Q9*

(a) $0.79 - 0.095 = \boxed{0.695}$ (b)(i) $\boxed{z = -0.806}$ (ii) $\frac{289-297}{\sigma} = -0.8064 \Rightarrow \boxed{\sigma = 9.92}$

(c) 35th percentile $= 297 + 9.92(-0.3853) = 293.2$: $\boxed{293\text{ g}}$

(d) $\mathrm{B}(36, 0.65)$: $\mathrm{P}(X \geq 19) = \boxed{0.954}$ (e) $0.954^2 = \boxed{0.911}$

49

May 2015 · Paper 2 · TZ2 · Q9

A machine manufactures a large number of nails. The length, L mm, of a nail is normally distributed, where $L \sim \mathrm{N}(50, \sigma^2)$.

(a) Find $\mathrm{P}(50-\sigma < L < 50+2\sigma)$. [3]

(b) The probability that the length of a nail is less than $53.92\,\text{mm}$ is 0.975.
Show that $\sigma = 2.00$ (correct to three significant figures). [2]

All nails with length at least t mm are classified as large nails.

(c) A nail is chosen at random. The probability that it is a large nail is 0.75.
Find the value of t. [3]

(d) (i) A nail is chosen at random from the large nails. Find the probability that the length of this nail is less than $50.1\,\text{mm}$.

(ii) Ten nails are chosen at random from the large nails. Find the probability that at least two nails have a length that is less than $50.1\,\text{mm}$. [8]

49 *May 2015 · Paper 2 · TZ2 · Q9*

(a) $\mathrm{P}(-1 < Z < 2) = \boxed{0.819}$ (b) $\frac{53.92-50}{\sigma} = 1.95996 \Rightarrow \sigma = 2.00$ ✓

(c) $\mathrm{P}(L \geq t) = 0.75 \Rightarrow t = 50 - 2(0.6745) = \boxed{48.7}$

(d)(i) $\dfrac{\mathrm{P}(48.65 < L < 50.1)}{0.75} = \boxed{0.360}$ (ii) $\mathrm{B}(10, 0.360)$: $\mathrm{P}(\geq 2) = \boxed{0.924}$

50

May 2015 · Paper 2 · TZ1 · Q9

A company makes containers of yogurt. The volume of yogurt in the containers is normally distributed with a mean of $260\,\text{ml}$ and standard deviation of $6\,\text{ml}$.

A container which contains less than $250\,\text{ml}$ of yogurt is **underfilled**.

(a) A container is chosen at random. Find the probability that it is underfilled. [2]

The company decides that the probability of a container being underfilled should be reduced to 0.02. It decreases the standard deviation to σ and leaves the mean unchanged.

(b) Find σ. [4]

The company changes to the new standard deviation, σ, and leaves the mean unchanged. A container is chosen at random for inspection. It passes inspection if its volume of yogurt is between 250 and $271\,\text{ml}$.

(c) (i) Find the probability that it passes inspection.

(ii) Given that the container is **not** underfilled, find the probability that it passes inspection. [6]

(d) A sample of 50 containers is chosen at random. Find the probability that 48 or more of the containers pass inspection. [4]

50 *May 2015 · Paper 2 · TZ1 · Q9*

(a) $\mathrm{P}(V < 250) = \boxed{0.0478}$ (b) $\frac{250-260}{\sigma} = -2.0537 \Rightarrow \boxed{\sigma = 4.87}$

(c)(i) $\mathrm{P}(250 < V < 271) = \boxed{0.968}$ (ii) $\frac{0.968}{0.98} = \boxed{0.988}$ (d) $\mathrm{B}(50, 0.968)$:

$\mathrm{P}(\geq 48) = \boxed{0.786}$

51

May 2014 · Paper 2 · TZ2 · Q10

A forest has a large number of tall trees. The heights of the trees are normally distributed with a mean of 53 metres and a standard deviation of 8 metres. Trees are classified as giant trees if they are more than 60 metres tall.

(a) A tree is selected at random from the forest.

(i) Find the probability that this tree is a giant.

(ii) Given that this tree is a giant, find the probability that it is taller than 70 metres. *[6]*

(b) Two trees are selected at random. Find the probability that they are both giants. *[2]*

(c) 100 trees are selected at random.

(i) Find the expected number of these trees that are giants.

(ii) Find the probability that at least 25 of these trees are giants. *[6]*

51 *May 2014 · Paper 2 · TZ2 · Q10*

(a)(i) $\mathrm{P}(H > 60) = \boxed{0.191}$ (ii) $\dfrac{\mathrm{P}(H > 70)}{\mathrm{P}(H > 60)} = \boxed{0.0880}$ (b) $0.191^2 = \boxed{0.0364}$

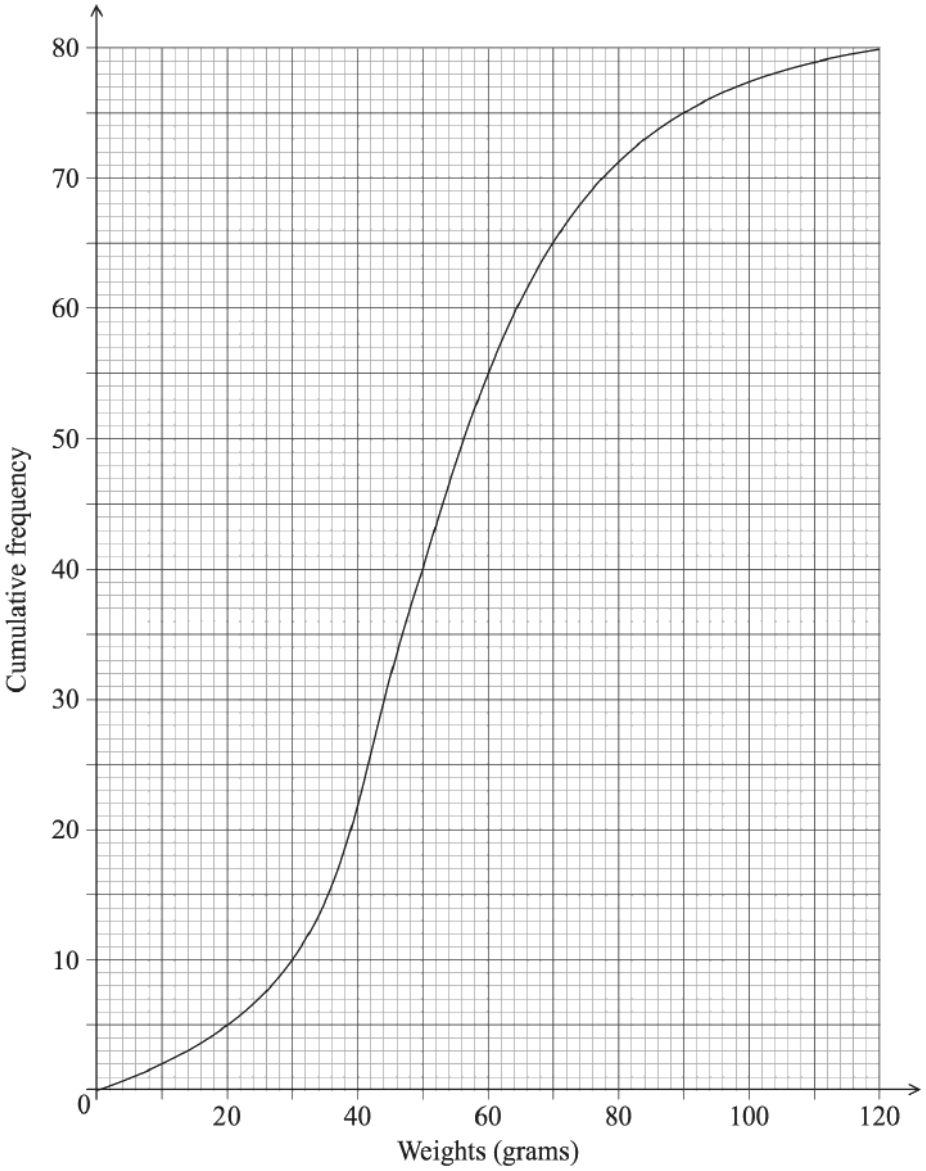
(c)(i) $100(0.191) = \boxed{19.1}$ (ii) $\mathrm{B}(100, 0.191)$: $\mathrm{P}(\geq 25) = \boxed{0.0869}$

52

May 2014 · Paper 2 · TZ1 · Q8

[Maximum mark: 16]

The weights in grams of 80 rats are shown in the following cumulative frequency diagram.



(a) (i) Write down the median weight of the rats.

(ii) Find the percentage of rats that weigh 70 grams or less. *[4]*

The same data is presented in the following table.

Weights w grams	$0 \le w \le 30$	$30 < w \le 60$	$60 < w \le 90$	$90 < w \le 120$
Frequency	p	45	q	5

Question continues on the next page →

(b) (i) Write down the value of p.

(ii) Find the value of q. *[4]*

(c) Use the values from the table to estimate the mean and standard deviation of the weights. *[3]*

Assume that the weights of these rats are normally distributed with the mean and standard deviation estimated in part (c).

(d) Find the percentage of rats that weigh 70 grams or less. *[2]*

(e) A sample of five rats is chosen at random. Find the probability that at most three rats weigh 70 grams or less. *[3]*

52 *May 2014 · Paper 2 · TZ1 · Q8*

(a)(i) cumulative frequency 40: $\boxed{50\text{ g}}$ (ii) about 65 rats: $\frac{65}{80} = \boxed{81\%}$ (accept 80–82%)

(b)(i) $\boxed{p = 10}$ (ii) $q = 80 - 10 - 45 - 5 = \boxed{20}$

(c) mid-points 15, 45, 75, 105: mean $= \boxed{52.5}$, s.d. $= \boxed{22.5}$

(d) $\mathrm{P}(W \leq 70) = \boxed{0.782}$ i.e. 78.2% (e) $\mathrm{B}(5, 0.782)$: $\mathrm{P}(\leq 3) = \boxed{0.301}$